B. MAIZE: CROP PROTECTION : SORGHUM DOWNY MILDEW

Season : Rabi- 2022 -2023

1. Experiment No. MPT-1(1099): Screening Of late Maturity Maize Entries Against Sorghum Downy Mildew			
1.	**Objective:** To identify resistant source against SDM		
2.	**Experimental Details :**		
a) Location : ZARS, Mandya	**b) Design :**Alpha	**c) No. of entries: 42** + 2	
d) Irrigation / Rainfed: Irrigated	**e) Replication :** Two	**f) Fertilizer (NPK kg/ha)** :150:75:40	
g) Spacing : 60 x 30 cm	**h) Plot Size:**1 Row X 2.5 m Length		
3.	**Date of Sowing: (SR) :**29.11.2022 & **(TE) :**28.12.2022	**4.**	**Date of inoculation :**03.12.2022 **Date of observation :**28.01.2023
5.	**Name of the susceptible check :**African tall		

1. **6. Results**

Sl. No.	Genotype	PDI (%)	Reaction	Sl. No.	Genotype	PDI (%)	Reaction
1	DM 9445	90.0	S	23	SMH 6222	38.6	MS
2	AHD 8721	100.0	S	24	SMH 6333	4.2	R
3	AHD 8722	38.7	MS	25	BH 417193	85.7	S
4	AH 8245	66.7	S	26	TMMH 2845	100.0	S
5	BS 650	100.0	S	27	ADV 7559	20.0	MR
6	KMH 8543	77.5	S	28	ADV 7415	32.1	MS
7	KMH 8585	65.8	S	29	PM21204L	73.2	S
8	KMH 8333	3.6	R	30	PM21207L	87.5	S
9	BH 417157	77.5	S	31	VNR 4597	68.9	S
10	Ellora Commando1112	57.9	S	32	TMMH 2882	9.3	R
11	ADV 7126	85.0	S	33	ADV 7733	60.0	S
12	ADV 7251	36.3	MS	34	PM20205L	100.0	S
13	MMH 22-62	65.2	S	35	NMH 713(C)	100.0	S
14	MMH 22-67	95.8	S	36	KHM 25K45(C)	100.0	S
15	PM 22201L	96.7	S	37	P 3522 (C)	100.0	S
16	PM 22202L	44.6	MS	38	CM 600 (C)	100.0	S
17	PM 22203L	86.4	S	39	BIO 9544 (C)	95.8	S
18	PM 22204L	100.0	S	40	Surya (C)	100.0	S
19	PM 22205L	100.0	S	41	African Tall(C)	100.0	S
20	PM 22206L	100.0	S	42	DHM 117(Filler)	100.0	S
21	PM 22207L	96.7	S	RC	NAH 1137	8.8	R
22	PM 22208L	91.7	S	SC	African Tall	90.3	S

Note: SR- Spreader rows; TE-Test entries; RC= Resistant check; SC=Susceptible check & NG- Not germinated.

Inference: In the evaluation of 42 entries, only two entries SMH 6333 and TMMH 2882 showed resistance to SDM, whereas ADV 7559 entry showed a moderately resistant reaction against SDM. Additionally, the entry NAH 1137 recorded a resistance level of 8.8% against SDM. On the other hand, the susceptible check, African Tall, exhibited a

susceptibility level of 90.3%, indicating its vulnerability to the disease. These results underscore the presence of promising resistant entries that could contribute to the development of maize varieties with enhanced resistance characteristics.

Scientists Involved: Drs.Jadesha. G, Swapna. G, Mallikarjuna. N and Mahadevu. P.

2. Experiment No. MPT-2(1098): Screening Of Medium Maturity Maize Entries Against Sorghum Downy Mildew			
1.	**Objective:** To identify resistant source against SDM		
2.	**Experimental Details :**		
a) Location : ZARS, Mandya	**b) Design :**Alpha	**c) No. of entries: 50** + 2	
d) Irrigation / Rainfed: Irrigated	**e) Replication :** Two	**f) Fertilizer (NPK kg/ha) :**150:75:40	
g) Spacing : 60 x 30 cm	**h) Plot Size:**1 Row X 2.5 m Length		
3.	**Date of Sowing: (SR) :**29.11.2022 & (TE) :28.12.2022	**4.**	**Date of inoculation :**03.12.2022 **Date of observation :**28.01.2023
5.	**Name of the susceptible check :**African tall		

6. Results

Sl. No.	Genotype	PDI (%)	Reaction	Sl. No.	Genotype	PDI (%)	Reaction
1	VAH 20001	100	S	27	MMH 22-52	100	S
2	AHD 2065	71.7	S	28	IMH 2-22R-12	96.9	S
3	AH 8194	80	S	29	BAUMH 21-1	96.4	S
4	AH 8728	20	MR	30	BAUMH 21-3	100	S
5	IMH 2-22R-1	100	S	31	BAUMH 21-4	92.5	S
6	IMH 2-22R-2	100	S	32	BRM19-1	86.7	S
7	IMH 2-22R-3	100	S	33	BRM 19-2	90	S
8	IMH 2-22R-4	89.8	S	34	IMHSB 21R-7	92.9	S
9	IMH 2-22R-5	93.8	S	35	IMHSB 20R-3	81.3	S
10	IMH 2-22R-6	67.5	S	36	IMHSB 20R-11	67.5	S
11	IMH 2-22R-7	78.6	S	37	SYN217806	100	S
12	IMH 2-22R-8	100	S	38	BMH-18-2	70	S
13	IMH 2-22R-9	96.4	S	39	BH 417206	55.2	S
14	IMH 2-22R-10	100	S	40	BH 417182	58.1	S
15	IMH 2-22R-11	83.3	S	41	IMHSB 20R-10	96.7	S
16	SYN 227808	100	S	42	IMHSB 20R-6	100	S
17	SYN 227809	100	S	43	SYN207660	100	S
18	SYN 227705	87.5	S	44	SYN207884	100	S
19	OMH 22R-2	81.3	S	45	DKC 8225	100	S
20	BS 652	54	S	46	BIO 9544(C)	100	S
21	BH 417167	92.9	S	47	DHM 117(C)	100	S
22	KMH 22-5	100	S	48	CM 600(C)	100	S
23	Ellora Virat-12	96.7	S	49	SURYA (C)	100	S
24	AH 4672	93.1	S	50	African Tall(C)	100	S
25	AH 4682	96.4	S	RC	NAH 1137	7.8	R
26	MMH 22-45	85.7	S	SC	African Tall	95.5	S

Note: SR- Spreader rows; TE-Test entries; RC= Resistant check; SC=Susceptible check & NG- Not germinated.

Inference : Out of 50 entries screened for Sorghum Downey mildew, none of the entry exhibited resistance to SDM. Whereas, only one entry AH 8728 showed moderately resistance reaction to SDM. Additionally, the entry NAH 1137 recorded a resistance level of 7.8% against SDM. On the other hand, the susceptible check, African Tall, exhibited a susceptibility level of 95.5%, indicating its vulnerability to the disease. These results underscore the presence of promising resistant entries that could contribute to the development of maize varieties with enhanced resistance characteristics.

Scientists Involved: Drs.Jadesha. G, Swapna. G, Mallikarjuna. N and Mahadevu. P.

3. Experiment No. MPT-3(1100): Screening Of Specialty corn Maize Entries Against Sorghum Downy Mildew			
1.	**Objective:** To identify resistant source against SDM		
2.	**Experimental Details :**		
a) Location : ZARS, Mandya	**b) Design :**Alpha	**c) No. of entries: 50** + 2	
d) Irrigation / Rainfed: Irrigated	**e) Replication :** Two	**f) Fertilizer (NPK kg/ha) :**150:75:40	
g) Spacing : 60 x 30 cm	**h) Plot Size:**1 Row X 2.5 m Length		
3.	**Date of Sowing: (SR) :**29.11.2022 & **(TE) :**28.12.2022	**4.**	**Date of inoculation :**03.12.2022 **Date of observation :**28.01.2023
5.	**Name of the susceptible check :**Afrioan tall		

6.Results:

Sl. No.	Genotype	PDI (%)	Reaction	Sl. No.	Genotype	PDI (%)	Reaction
1	APQH1	100	S	27	HPC 4	100	S
2	APQH5	100	S	28	HPC 5	100	S
3	APQH7	100	S	29	IMHP 2101	100	S
4	IQH7-219	100	S	30	MPCH 22-22	95.8	S
5	IQH7-215	85	S	31	MPC 2141	100	S
6	QPMMH 22-35	95	S	32	PP 09	100	S
7	MQH 21-51	95	S	33	PP II	100	S
8	LQMH1(C)	96.4	S	34	LPCH3 (C)	96.4	S
9	HQPM1 (C)	95	S	35	DMRHP 1402(C)	100	S
10	PUSA BIOFORTIFIED MAIZE HYBRID-1(C)	100	S	36	BPCH6 (C)	100	S
11	HQPM-5(C)	100	S	37	IIVRBCH-1613	96.7	S
12	HQPM-7(C)	100	S	38	IBH 9-224	96.7	S
13	BSCH 418184	100	S	39	IBH9-225	100	S
14	HI BRIX 33	86.7	S	40	MBC 22-32	70	S
15	HI BRIX 59	73.6	S	41	MBC 21-10	91.7	S
16	HI BRIX 81	96.4	S	42	BAUMH 21-8	100	S
17	MSCH 22- 28	100	S	43	BAUMH 21-1-1	100	S
18	MISHTI(C)	100	S	44	AH 7043(C)	100	S

19	ASKH 4(C)	9.3	R	45	ABSH-1(C)	100	S
20	VL Sweet Corn(C)	100	S	46	CMVL Baby corn 1(C)	100	S
21	CHCP 2204	100	S	47	BIO 9544(C)	95	S
22	CHCP 2205	100	S	48	Surya (C)	100	S
23	CHCP 2206	100	S	49	CM 600(C)	100	S
24	CHCP 2207	95	S	50	African Tall (C)	100	S
25	APCH 4	100	S	RC	NAH 1137	8.8	R
26	IPH 2-22R-1	100	S	SC	African Tall	96.5	S

Note: SR- Spreader rows; TE-Test entries; RC= Resistant check; SC=Susceptible check & NG- Not germinated.

Inference : Among the evaluated 50 entries of specialty corn, only one entry, ASKH 4(C) showed resistance to SDM. The susceptible check, African Tall, demonstrated a susceptibility level of 95.5%, indicating its vulnerability to the disease. In contrast, the resistant check NAH 1137 recorded a resistance level of 7.8% against SDM. These findings highlight the absence of resistance in the evaluated specialty corn entries and emphasize the importance of incorporating resistant checks like MAH-14-138 in breeding programs for enhanced disease resistance.

Scientists Involved: Drs.Jadesha. G, Swapna. G, Mallikarjuna. N and Mahadevu. P.

4. Experiment No.: Screening of Station inbred Maize Entries Against Sorghum Downy Mildew			
1.	**Objective:** To identify resistant source against SDM		
2.	**Experimental Details :**		
a) Location : ZARS, Mandya	**b) Design :**Alpha	**c) No. of entries: 38** + 2	
d) Irrigation / Rainfed: Irrigated	**e) Replication :** Two	**f) Fertilizer (NPK kg/ha) :**150:75:40	
g) Spacing : 60 x 30 cm	**h) Plot Size:**1 Row X 2.5 m Length		
3.	**Date of Sowing: (SR) :**29.11.2022 & **(TE) :**28.12.2022	**4.**	**Date of inoculation :**03.12.2022 **Date of observation :**28.01.2023
5.	**Name of the susceptible check :**African tall		

6.Results:

Sl. No.	Genotype	PDI (%)	Reaction	Sl. No.	Genotype	PDI (%)	Reaction
1	MAI-1	44.4	MS	21	NAI-137-6	58.7	S
2	MAI-2	28.09	MS	22	NAI-137-7	0	R
3	MAI-3	56	S	23	MAI-204-1	26.8	MS
4	MAI-7	71.1	S	24	CAL-1443	18.75	MR
5	MAI-8	33.75	MS	25	CML-451	0	R
6	MAI-10	100	S	26	1298	96.15	S
7	MAI-10-1	7.14	R	27	32129	60.6	S
8	MAI-12	63.3	S	28	V-496-7	NG	NG
9	MAI-13	50.7	S	29	40402	90.9	S
10	MAI-12	35.7	MS	30	52067	71.6	S
11	MAI-20	79.1	S	31	QPM 25	20	MR

12	MAI-142	100	S	32	QPM 26	36.1	MS
13	NAI-204-1	NG	NG	33	QPM 26-1	95	S
14	MAI 215	13.3	R	34	MAI-729	0	R
15	MAI-215-1	0	R	35	SKV 50	50	MS
16	MAI-224	19.6	MR	36	V-940-31	88.8	S
17	MAI 276	32.5	MS	37	52336	21.4	MR
18	NAI-137	52.2	S	38	MAI-105	0	R
19	MAI 753	12.5	MR	RC	NAH 1137	9.8	R
20	MAI 755	41.4	MS	SC	African Tall	100.0	S

Note: SR- Spreader rows; TE-Test entries; RC= Resistant check; SC=Susceptible check & NG- Not germinated.

Inference: Among the 38 maize station inbreds, entries MAI-10-1, MAI 215, MAI-215-1, NAI-137-7, CML-451, MAI-729, and MAI-105 exhibited resistance to SDM, while entries MAI-224, MAI 753, CAL-1443, QPM 25, and 52336 were found to be moderately resistant. The resistant check, NAH 1137, recorded a resistance level of 9.8%, indicating its effectiveness against SDM. Additionally, the susceptible check, African Tall, showed complete susceptibility with a 100% disease incidence. These results provide insights into the resistance levels of the evaluated maize station inbreds, with NAH 1137 serving as a promising resistant check.

Scientists Involved: Drs.Jadesha. G, Swapna. G, Mallikarjuna. N and Mahadevu. P.

5. Experiment No. : Screening of Station Hybrids and their Parents Maize Entries Against Sorghum Downy Mildew			
1.	**Objective:** To identify resistant source against SDM		
2.	**Experimental Details :**		
a) Location : ZARS, Mandya	**b) Design :**Alpha	**c) No. of entries: 21** + 2	
d) Irrigation / Rainfed: Irrigated	**e) Replication :** Two	**f) Fertilizer (NPK kg/ha) :**150:75:40	
g) Spacing : 60 x 30 cm	**h) Plot Size:**1 Row X 2.5 m Length		
3.	**Date of Sowing: (SR) :**29.11.2022 & **(TE) :**28.12.2022	**4.**	**Date of inoculation :**03.12.2022 **Date of observation :**28.01.2023
5.	**Name of the susceptible check :**African tall		

6.Results:

Sl. No.	Genotype	PDI (%)	Reaction	Sl. No.	Genotype	PDI (%)	Reaction
1	SKV-50-6	0	R	13	NAH- 2049 II	10	R
2	MAH-22-2	90	S	14	CAL-1443	10.7	MR
3	NAH-1137	41	MS	15	MAI-215	6.25	R
4	MAH-22-4	100	S	16	MAH-19-2	10	R
5	MAH-14-5	7.14	R	17	SKV-50 TALL	0	R
6	MAH-22-1	95.8	S	18	MAI-105	0	R
7	NAI-137	86.3	S	19	MAI-729	0	R
8	CML-581	25	MR	20	CML-582	38.6	MS

9	NAH-2049- I	9.4	R	21	MAH-14-138	0	R
10	MAH-22-5	100	S	RC	NAH-1137	6.5	R
11	MAH-22-3	100	S	SC	African Tall	98.7	S
12	CML-451	8.3	R				

Note: SR- Spreader rows; TE-Test entries; RC= Resistant check; SC=Susceptible check & NG- Not germinated.

Inference: In the screening of 21 station hybrids and their parents against Sorghum Downy Mildew, entries SKV-50-6, MAH-14-5, NAH-2049-I, CML-451, NAH-2049-II, MAI-215, MAH-19-2, SKV-50 TALL, MAI-105, MAI-729, and MAH-14-138 exhibited resistance, while entries CML-581 and CAL-1443 were found to be moderately resistant. The resistant check, NAH-1137, recorded a resistance level of 6.5%, indicating its effectiveness against Sorghum Downy Mildew. Additionally, the susceptible check, African Tall, showed high susceptibility with a 98.7% disease incidence. These findings highlight the potential resistance of certain entries, including the effectiveness of NAH-1137, in combating Sorghum Downy Mildew.

Scientists Involved: Drs.Jadesha. G, Swapna. G, Mallikarjuna. N and Mahadevu. P.

6. Experiment No. : Screening of IIMR-WNC maize inbreds Against Sorghum Downy Mildew			
1.	**Objective:** To identify resistant source against SDM		
2.	**Experimental Details :**		
a) Location : ZARS, Mandya	**b) Design :**Alpha	**c) No. of entries: 100** + 2	
d) Irrigation / Rainfed: Irrigated	**e) Replication :** Two	**f) Fertilizer (NPK kg/ha) :**150:75:40	
g) Spacing : 60 x 30 cm	**h) Plot Size:**1 Row X 2.5 m Length		
3.	**Date of Sowing: (SR) :**29.11.2022 & **(TE) :**28.12.2022	**4.**	**Date of inoculation :**03.12.2022 **Date of observation :**28.01.2023
5.	**Name of the susceptible check :**African tall		

2. 6.Results:

Sl. No.	Genotype	PDI (%)	Reaction	Sl. No.	Genotype	PDI (%)	Reaction
1	109463	90.9	S	52	109850	80.0	S
2	109466	76.9	S	53	109860	100.0	S
3	109473	100.0	S	54	109864	100.0	S
4	109484	100.0	S	55	109866	92.9	S
5	109493	37.5	MS	56	109868	14.3	MR
6	109497	93.3	S	57	109869	100.0	S
7	109514	84.6	S	58	109870	30.0	MS
8	109526	100.0	S	59	109872	100.0	S
9	109530	100.0	S	60	109876	100.0	S
10	109533	100.0	S	61	109879	37.5	MS
11	109534	46.2	MS	62	109880	76.9	S
12	109544	100.0	S	63	109888	100.0	S
13	109550	100.0	S	64	109910	100.0	S
14	109580	35.3	MS	65	109924	85.7	S
15	109583	46.2	MS	66	109934	41.7	MS

16	109614	100.0	S	67	109935	80.0	S
17	109619	90.9	S	68	109936	100.0	S
18	109642	92.3	S	69	109937	100.0	S
19	109646	7.7	R	70	109939	100.0	S
20	109654	100.0	S	71	109940	100.0	S
21	109657	100.0	S	72	109942	37.5	MS
22	109659	100.0	S	73	109947	100.0	S
23	109661	31.3	MS	74	109948	100.0	S
24	109667	92.9	S	75	109952	91.7	S
25	109688	100.0	S	76	109954	83.3	S
26	109690	100.0	S	77	109964	61.1	S
27	109702	100.0	S	78	109971	100.0	S
28	109710	100.0	S	79	109992	100.0	S
29	109712	100.0	S	80	109994	0.0	R
30	109715	43.8	MS	81	109996	100.0	S
31	109718	100.0	S	82	109998	100.0	S
32	109719	100.0	S	83	110001	100.0	S
33	109726	100.0	S	84	110007	100.0	S
34	109727	62.5	S	85	110008	100.0	S
35	109728	90.9	S	86	110012	46.2	MS
36	109729	100.0	S	87	110013	53.3	S
37	109732	100.0	S	88	110014	41.7	MS
38	109736	100.0	S	89	110020	40.0	MS
39	109743	100.0	S	90	110064	44.4	MS
40	109747	100.0	S	91	110065	62.5	S
41	109753	100.0	S	92	110067	22.2	MR
42	109754	100.0	S	93	110071	100.0	S
43	109761	100.0	S	94	110076	100.0	S
44	109765	100.0	S	95	110114	100.0	S
45	109773	100.0	S	96	110130	40.0	MS
46	109774	100.0	S	97	110134	100.0	S
47	109775	83.3	S	98	110137	100.0	S
48	140977	100.0	S	99	110141	33.3	MS
49	109824	100.0	S	100	110154	100.0	S
50	109844	100.0	S	RC	NAH-1137	6.5	R
51	109845	25.0	MR	SC	African Tall	98.7	S

Note: SR- Spreader rows; TE-Test entries; RC= Resistant check; SC=Susceptible check & NG- Not germinated.

Inference : Among the 100 IIMR WNC maize entries, entries 109646 and 109994 exhibited resistance, while entries 109845, 109868, and 110067 were found to be moderately resistant against the disease. The resistant check, NAH-1137, demonstrated a resistance level of 6.5%, signifying its efficacy in combating the disease. On the other hand, the susceptible check, African Tall, displayed a high susceptibility with a disease incidence of 98.7%. These results

emphasize the presence of resistant and moderately resistant entries in the IIMR WNC maize collection.

Scientists Involved: Drs.Jadesha. G, Swapna. G, Mallikarjuna. N and Mahadevu. P.

7. Experiment No. : Screening of CIMMYT lines Against Sorghum Downy Mildew				
1.	**Objective:** To identify resistant source against SDM			
2.	**Experimental Details :**			
a) Location : ZARS, Mandya		**b) Design :**Alpha	**c) No. of entries: 42** + 2	
d) Irrigation / Rainfed: Irrigated		**e) Replication :** Two	**f) Fertilizer (NPK kg/ha) :**150:75:40	
g) Spacing : 60 x 30 cm		**h) Plot Size:**1 Row X 2.5 m Length		
3.	**Date of Sowing: (SR) :**29.11.2022 & **(TE) :**28.12.2022	**4.**	**Date of inoculation :**03.12.2022 **Date of observation :**28.01.2023	
5.	**Name of the susceptible check :**African tall			

6.Results:

Sl. No.	Genotype	PDI (%)	Reaction	Sl. No.	Genotype	PDI (%)	Reaction
1	VL-19978	10	R	23	VL-18260	0	R
2	VL-191001	10	R	24	VL-22461	0	R
3	VL-181520	0	R	25	VL-21966	0	R
4	VL-21986	0	R	26	VL-21988	50	MS
5	VL-185586	0	R	27	VL-184578	0	R
6	VL-18722	0	R	28	VL-18726	0	R
7	VL-18730	12.5	MR	29	VL-22294	0	R
8	VL-22302	0	R	30	VL-22327	5	R
9	VL-22308	0	R	31	SNL-18965	0	R
10	ZL-18869	0	R	32	SNL-20158	0	R
11	ZL-17578	0	R	33	SNL-172494	0	R
12	VL-181418	0	R	34	SNL-19388	0	R
13	VL-21927	0	R	35	SNL-20640	0	R
14	VL-18708	0	R	36	SNL-191161	0	R
15	VL-133728	0	R	37	SNL-191172	0	R
16	VL-20227	0	R	38	SNL-19123	0	R
17	VL-1110501	0	R	39	VL-181545	0	R
18	VL-191073	0	R	40	VL-181763	0	R
19	VL-18560	0	R	41	VL-19957	0	R
20	VL-22435	0	R	42	VL-18322	0	R
21	VL-22452	0	R	RC	NAH-1137	5.5	R
22	VL-183003	0	R	SC	African Tall	89.5	S

Note: SR- Spreader rows; TE-Test entries; RC= Resistant check; SC=Susceptible check & NG- Not germinated.

Inference : Among the 42 station CIMMYT maize lines, 40 entries displayed resistance to the disease. Additionally, one entry, VL-18730, exhibited moderate resistance, while another entry, VL-21988, showed moderate susceptibility. The resistant check, NAH-1137, recorded a

resistance level of 5.5%, confirming its effectiveness against the disease. Conversely, the susceptible check, African Tall, demonstrated a susceptibility level of 89.5%. These findings highlight the presence of resistant, moderately resistant, and moderately susceptible entries within the CIMMYT maize collection.

Scientists Involved: Drs.Jadesha. G, Swapna. G, Mallikarjuna. N and Mahadevu. P.

8. Experiment No. : Screening Of CRPAB Entries Against Sorghum Downy Mildew				
1.	**Objective:** To identify resistant source against SDM			
2.	**Experimental Details :**			
a) Location : ZARS, Mandya		**b) Design :**Alpha	**c) No. of entries: 197**+ 2	
d) Irrigation / Rainfed: Irrigated		**e) Replication :** Two	**f) Fertilizer (NPK kg/ha) :**150:75:40	
g) **Spacing :** 60 x 30 cm		**h) Plot Size:**1 Row X 2.5 m Length		
3.	**Date of Sowing: (SR) :**29.11.2022 & **(TE) :**28.12.2022		**4.**	**Date of inoculation :**03.12.2022 **Date of observation :**28.01.2023
5.	**Name of the susceptible check :**African tall			

6.Results:

Sl. No.	Genotype	PDI (%)	Reaction	Sl. No.	Genotype	PDI (%)	Reaction
1	791	40.0	MS	101	316	100.0	S
2	171	66.7	S	102	465	11.1	R
3	822	85.7	S	103	413	71.4	S
4	91	100.0	S	104	241	0.0	R
5	516	83.3	S	105	856	70.0	S
6	950	87.5	S	106	569	80.0	S
7	187	28.6	MS	107	689	100.0	S
8	829	80.0	S	108	835	40.0	MS
9	835	87.5	S	109	293	66.7	S
10	412	90.9	S	110	846	42.9	MS
11	586	80.0	S	111	870	80.0	S
12	53	90.0	S	112	457	37.5	MS
13	772	80.0	S	113	1033	80.0	S
14	717	80.0	S	114	586	100.0	S
15	37	100.0	S	115	798	100.0	S
16	842	66.7	S	116	993	40.0	MS
17	395	30.0	MS	117	815	100.0	S
18	17	83.3	S	118	334	70.0	S
19	553	0.0	R	119	271	33.3	MS
20	1033	100.0	S	120	1017	100.0	S
21	485	88.9	S	121	655	40.0	MS
22	569	0.0	R	122	346	100.0	S
23	723	85.7	S	123	82	30.0	MS
24	334	37.5	MS	124	789	90.0	S
25	411	66.7	S	125	508	25.0	MR

26	643	18.2	MR	126	822	100.0	S
27	1030	0.0	R	127	393	0.0	R
28	323	100.0	S	128	772	25.0	MR
29	52	0.0	R	129	787	50.0	MS
30	1005	0.0	R	130	43	25.0	MR
31	346	28.6	MS	131	411	25.0	MR
32	929	75.0	S	132	829	80.0	S
33	654	55.6	S	133	813	100.0	S
34	870	62.5	S	134	935	100.0	S
35	935	66.7	S	135	812	100.0	S
36	130	40.0	MS	136	17	25.0	MR
37	981	12.5	MR	137	323	57.1	S
38	895	44.4	MS	138	467	50.0	MS
39	832	14.3	MR	139	557	100.0	S
40	856	44.4	MS	140	543	75.0	S
41	769	25.0	MR	141	443	30.0	MS
42	246	40.0	MS	142	125	66.7	S
43	293	80.0	S	143	485	100.0	S
44	112	30.0	MS	144	734	40.0	MS
45	543	50.0	MS	145	336	53.8	S
46	510	0.0	R	146	178	0.0	R
47	413	87.5	S	147	855	100.0	S
48	813	100.0	S	148	769	100.0	S
49	457	62.5	S	149	950	11.1	MR
50	178	85.7	S	150	717	28.6	MS
51	443	90.0	S	151	19	0.0	R
52	997	50.0	MS	152	470	33.3	MS
53	488	60.0	S	153	981	55.6	S
54	82	60.0	S	154	516	0.0	R
55	655	28.6	MS	155	713	25.0	MR
56	70	83.3	S	156	896	55.6	S
57	465	100.0	S	157	70	37.5	MS
58	1028	66.7	S	158	37	50.0	MS
59	812	66.7	S	159	867	50.0	MS
60	674	33.3	MS	160	723	100.0	S
61	862	62.5	S	161	654	100.0	S
62	855	100.0	S	162	842	100.0	S
63	987	37.5	MS	163	791	71.4	S
64	734	100.0	S	164	839	55.6	S
65	548	40.0	MS	165	488	71.4	S
66	815	100.0	S	166	130	12.5	MR
67	816	85.7	S	167	548	60.0	S
68	13	100.0	S	168	1030	50.0	MS
69	787	50.0	MS	169	1039	55.6	S
70	634	66.7	S	170	987	100.0	S

71	789	100.0	S	171	1028	25.0	MR
72	467	100.0	S	172	997	50.0	MS
73	43	22.2	MS	173	696	NG	NG
74	711	44.4	MS	174	52	14.3	MR
75	689	50.0	MS	175	895	10.0	R
76	1017	100.0	S	176	13	100.0	S
77	846	33.3	MS	177	643	90.0	S
78	839	85.7	S	178	395	66.7	S
79	867	22.2	MR	179	91	33.3	MS
80	86	11.1	MR	180	960	75.0	S
81	19	100.0	S	181	929	71.4	S
82	760	62.5	S	182	53	20.0	MR
83	336	14.3	MR	183	86	36.4	MS
84	713	100.0	S	184	187	28.6	MS
85	241	0.0	R	185	510	50.0	MS
86	993	50.0	MS	186	684	10.0	R
87	557	0.0	R	187	112	20.0	MR
88	393	100.0	S	188	862	16.7	MR
89	508	0.0	R	189	831	50.0	MS
90	831	37.5	MS	190	711	44.4	MS
91	1039	42.9	MS	191	1005	30.0	MS
92	696	42.9	MS	192	674	27.3	MS
93	798	33.3	MS	193	412	100.0	S
94	125	100.0	S	194	171	22.2	MR
95	684	16.7	MR	195	832	27.3	MS
96	746	14.3	MR	196	746	25.0	MR
97	316	100.0	S	197	553	12.5	MR
98	470	50.0	MS	RC	NAH-1137	7.2	R
99	271	0.0	R	SC	African Tall	99.0	S
100	634	55.6	MS				

Note: SR- Spreader rows; TE-Test entries; RC= Resistant check; SC=Susceptible check & NG- Not germinated.

Inference : In the screening of 197 CRPAB entries against Sorghum Downy Mildew (SDM), a total of 18 entries, specifically, 553, 569, 1030, 52, 1005, 510, 241, 557, 508, 271, 465, 241, 393, 178, 19, 516, 895, and 684, demonstrated resistance (R) to SDM. Additionally, 24 entries, namely 643, 981, 832, 769, 867, 86, 336, 684, 746, 508, 772, 43, 411, 17, 950, 713, 130, 1028, 52, 53, 112, 862, 171, 746, and 553, exhibited moderate resistance (MR). The resistant check, NAH-1137, recorded a resistance level of 7.2%, confirming its efficacy against SDM. In contrast, the susceptible check, African Tall, displayed a susceptibility level of 99.0%. These results emphasize the presence of promising resistance in CRPAB entries against SDM.

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